

icant manual dexterity, precision, and personal interaction involved in barbering are challenging to replicate with current or medium-term advances in technology.

Tasks Nature (Routine vs Non-Routine). The model consistently distinguishes between routine and non-routine tasks. Occupations with routine, repetitive tasks are considered more likely to be replaced by AI. The term "routine" appears 72 times in the responses, indicating a strong emphasis on the predictability and repetitiveness of tasks. For example, the response for "Anesthesiologists" (likelihood 0.2) notes that although AI might assist with some routine tasks such as diagnostic procedures, monitoring patients, and scheduling, the holistic and critical (non-routine) nature of anesthesiology, including patient care, emergency response, coordination with medical teams, and individualized decision-making, presents substantial challenges to full automation.

Ethical Considerations. The model frequently raises ethical concerns, particularly for occupations involving direct human interaction or safety-critical roles. The terms "ethical" and "safety" appear 28 and 81 times, respectively, in the responses. For example, in the case of "Judges," ChatGPT notes that the complex and highly ethical nature of judicial tasks, such as interpreting laws and making nuanced judgments, poses significant challenges for AI, making it unlikely that judges will be replaced by AI soon (likelihood of 0.2). Similarly, in occupations like Pilots and Amusement and Recreation Attendants, safety concerns are raised to lower AI prospects of replacement.

Regulatory Environment. Legal and regulatory factors are also considered. Regulatory concerns were raised 63 times in the follow-up responses. This occurs in occupations like bakers, drivers, and civil engineers. Heavily regulated industries might be less likely due to strict regulations and standards that AI must meet.

Cost-Benefit Analysis. Economic feasibility also plays a role. For instance, jobs that require high (low) levels of human expertise and are less (more) economically viable for automation may have lower (higher) likelihoods. This factor is considered in occupations such as "Farm Equipment Mechanics and Service Technicians" and "Hand Packers and Packagers."

These findings not only provide insight into the model's decision-making process but also align with established theories in labor economics and technological change. The consideration of task routineness, for instance, echoes the task-based approach to technological change (Autor *et al.*, 2003, 2006; Acemoglu and Autor, 2011; Autor, 2013), while the emphasis on ethical and regulatory factors reflects the growing awareness of AI's societal implications.

3.2 Comparison with Existing Indexes

Figure 2 compares the GENOE index with other indexes of computerization and occupational exposure to technological displacement.⁶ Interestingly, our index shows little correlation with

⁶Since most indexes were calculated using the SOC-2010 classification, we transformed the GENOE from SOC-2018 to SOC-2010 using the crosswalks from the Bureau of Labor Statistics for comparison purposes. Henceforth, in this document, whenever the generated indexes are compared to existing indexes, they are transformed to SOC-2010.